Geostatistical Analysis of Ozone Concentrations in California

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```r
library(sp)
library(gstat)
library(geoR)
```

```
## --------------------------------------------------------------
##  Analysis of Geostatistical Data
##  For an Introduction to geoR go to http://www.leg.ufpr.br/geoR
##  geoR version 1.9-6 (built on 2025-08-29) is now loaded
## --------------------------------------------------------------
```

```r
library(ggplot2)
library(maps)

california <- map_data("state")
california <- subset(california, region == "california")
```

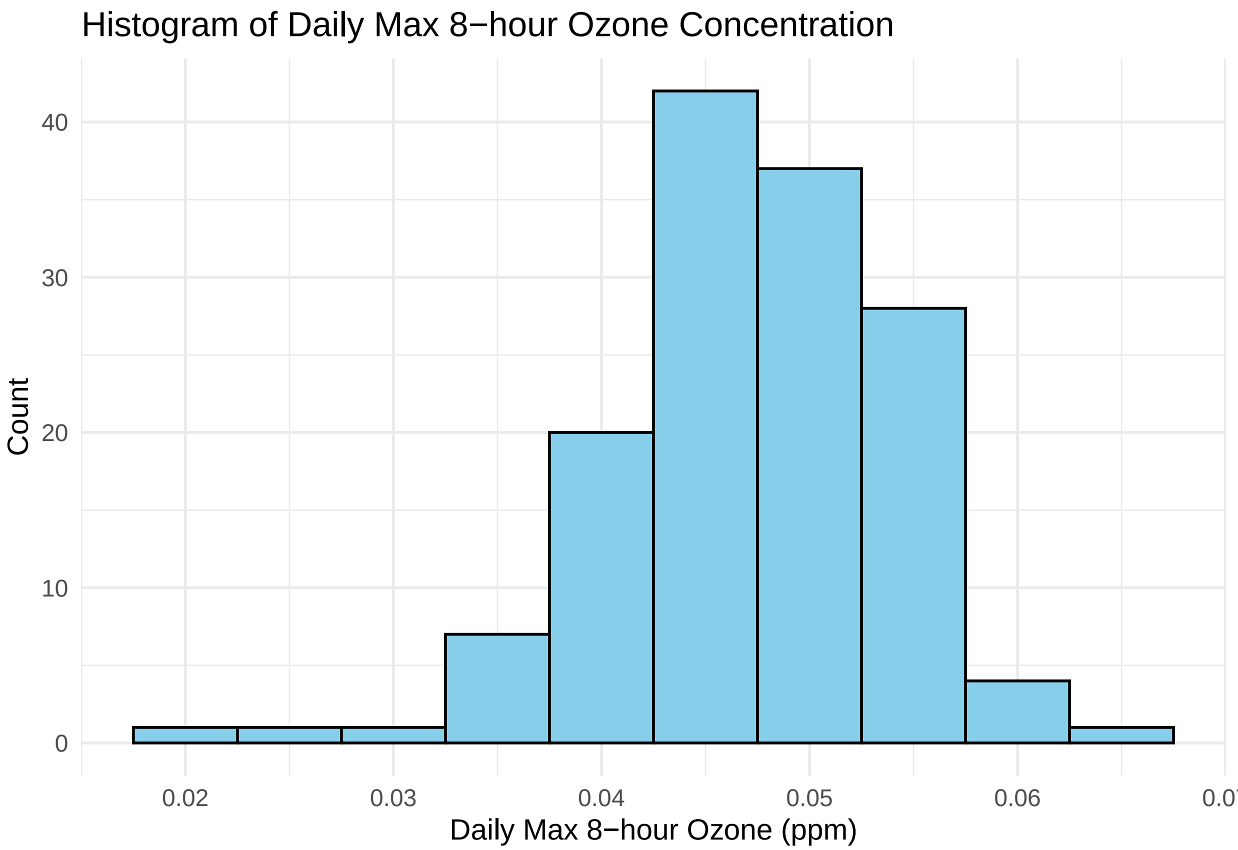
```r
data <- read.csv("Records.csv")

relevant <- data[data$Date == "03/10/2025", ]

mean(relevant$Daily.Max.8.hour.Ozone.Concentration)
```

```
## [1] 0.0471338
```

This data was collected at 142 unique locations across the state of California. The specific day that will be looked at 3/10/2025. The variable looked will be maximum of ozone concentartion over a period of any 8 consecutive hours in a day. The variable ranges from 0.022 to 0.066 with a mean 0.0471338.

Histogram

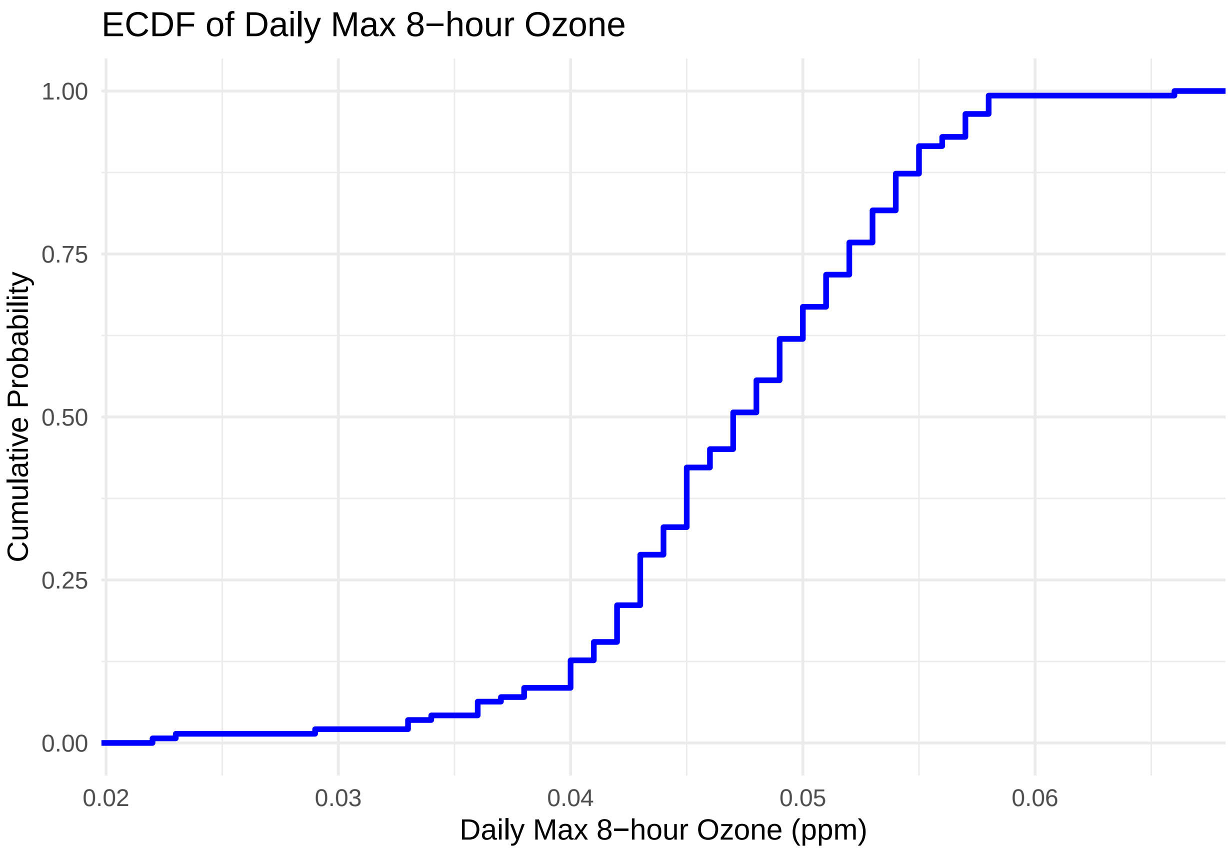
```r
ggplot(relevant, aes(x = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_histogram(binwidth = 0.005,
                 fill = "skyblue",
                 color = "black") +
  labs(
    title = "Histogram of Daily Max 8-hour Ozone Concentration",
    x = "Daily Max 8-hour Ozone (ppm)",
    y = "Count"
  ) +
  theme_minimal()
```



ECDF

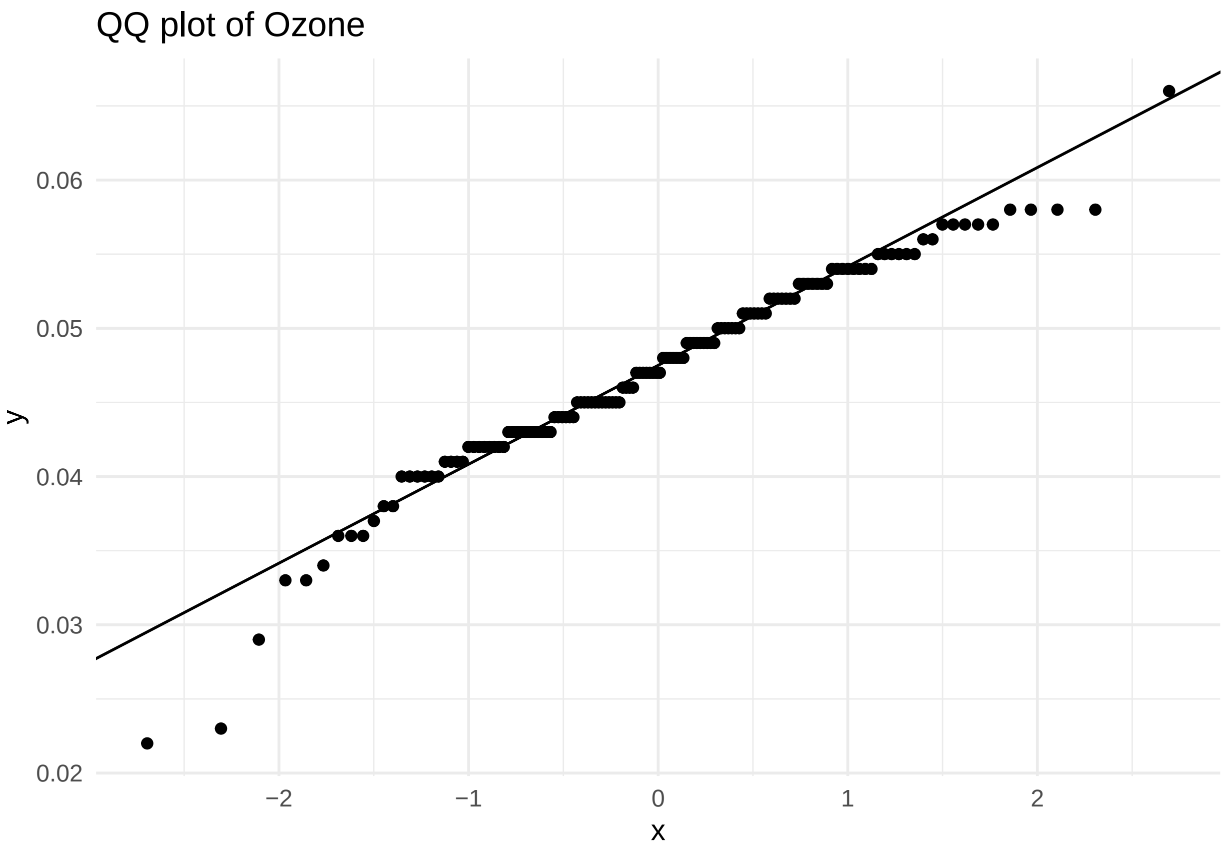
```r
ggplot(relevant, aes(x = Daily.Max.8.hour.Ozone.Concentration)) +
  stat_ecdf(geom = "step", color = "blue", size = 1) +
  labs(
    title = "ECDF of Daily Max 8-hour Ozone",
    x = "Daily Max 8-hour Ozone (ppm)",
    y = "Cumulative Probability"
  ) +
  theme_minimal()
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



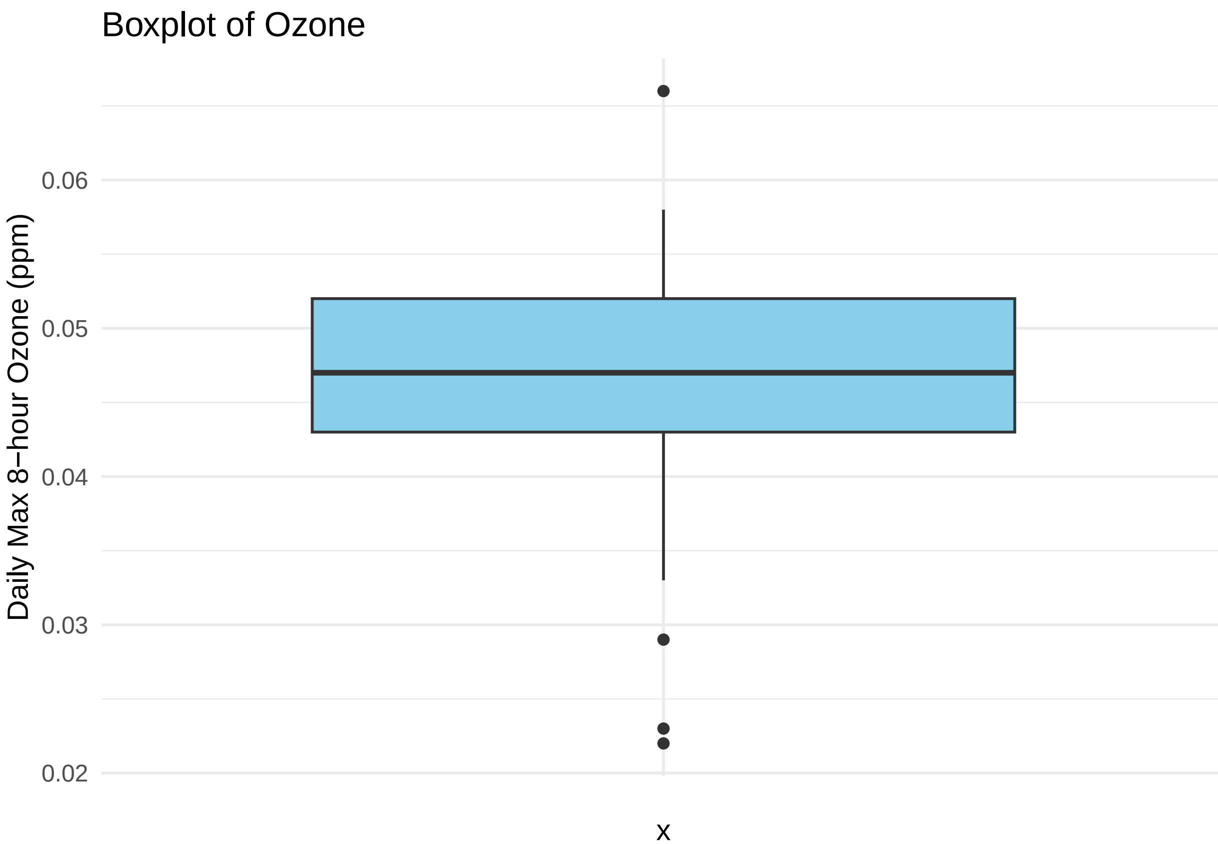
QQ Plot

```
ggplot(relevant, aes(sample = Daily.Max.8.hour.Ozone.Concentration)) +
  stat_qq() +
  stat_qq_line() +
  theme_minimal() +
  labs(title = "QQ plot of Ozone")
```



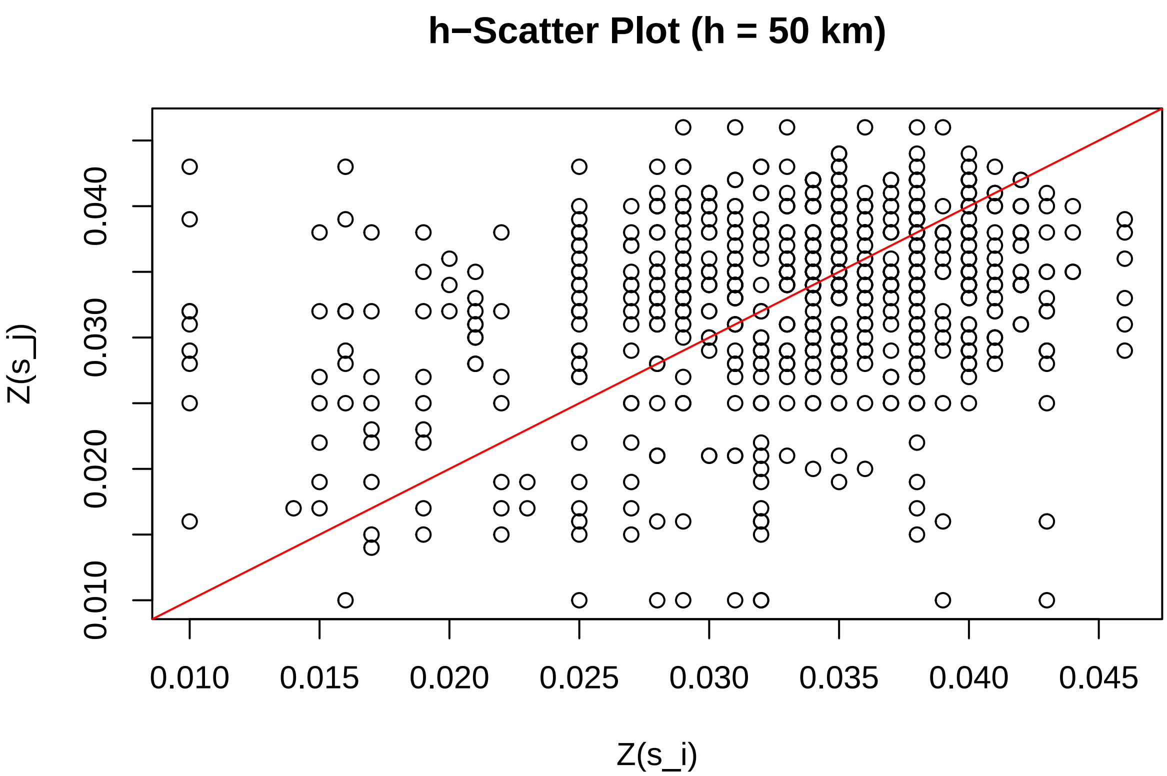
Boxplot

```
ggplot(relevant, aes(x = "", y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_boxplot(fill = "skyblue") +
  labs(y = "Daily Max 8-hour Ozone (ppm)", title = "Boxplot of Ozone") +
  theme_minimal()
```



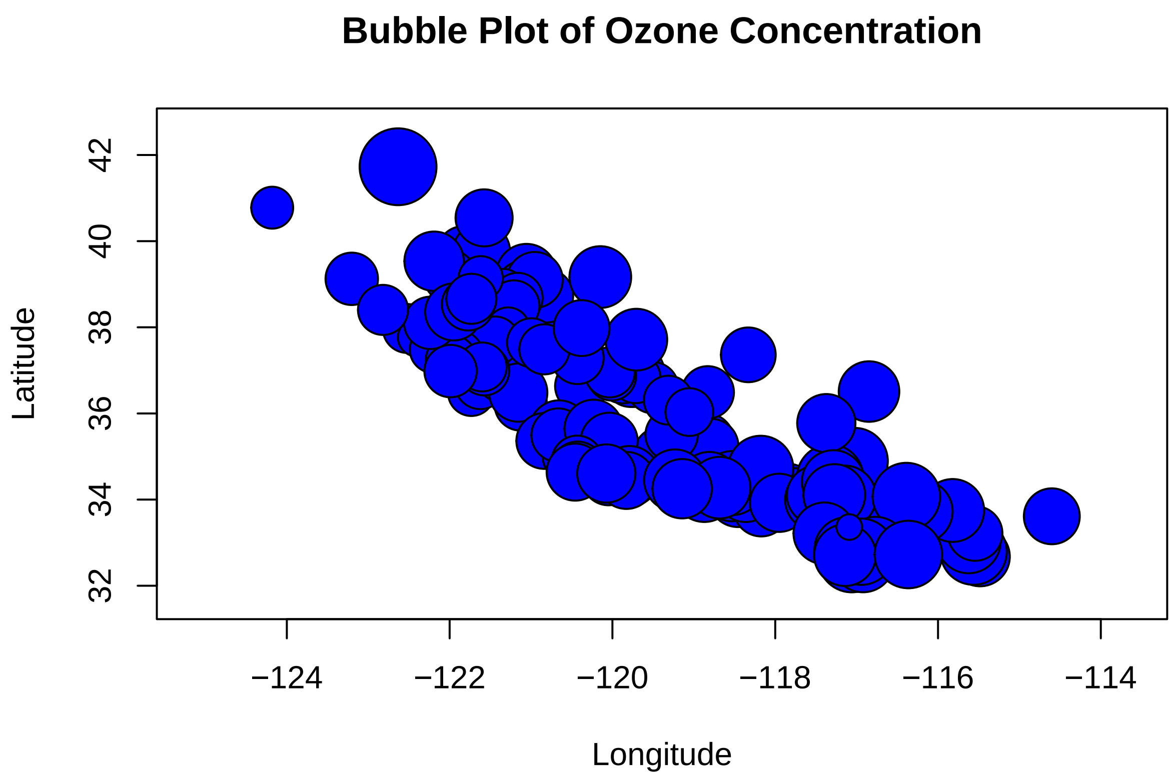
H Scatterplot

```
coordinates(relevant) <- ~Site.Longitude + Site.Latitude

dist_matrix <- spDists(coordinates(relevant))

ozone <- data$Daily.Max.8.hour.Ozone.Concentration

pairs <- which(dist_matrix > 0 & dist_matrix < 0.5, arr.ind = TRUE)

plot(
  ozone[pairs[,1]],
  ozone[pairs[,2]],
  xlab = "Z(s_i)",
  ylab = "Z(s_j)",
  main = "h-Scatter Plot (h = 50 km)"
)

abline(0,1,col="red")
```



Bubble Plot

```
symbols(
  relevant$Site.Longitude,
  relevant$Site.Latitude,
  circles = relevant$Daily.Max.8.hour.Ozone.Concentration,
  inches = 0.2,
  xlab = "Longitude",
  ylab = "Latitude",
  main = "Bubble Plot of Ozone Concentration",
  bg = "blue",
  fg = "black"
)
```



Variograms

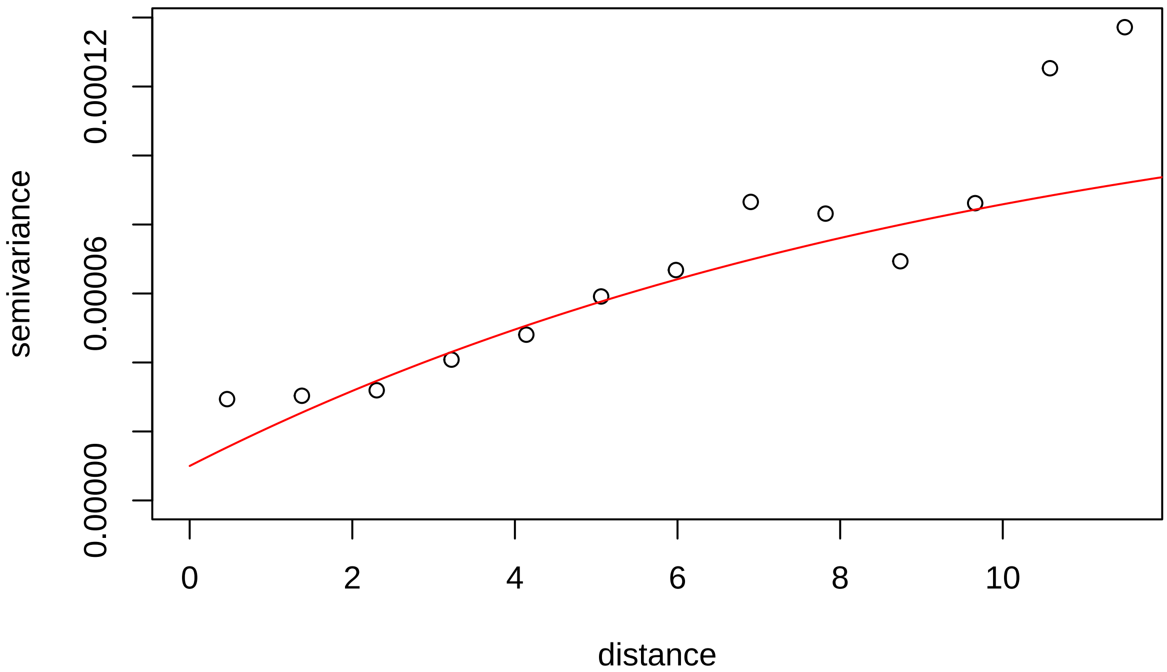
```
relevant <- as.data.frame(relevant)

ozone_geo <- as.geodata(
  relevant,
  coords.col = c("Site.Longitude","Site.Latitude"),
  data.col = "Daily.Max.8.hour.Ozone.Concentration"
)

var_classical <- variog(ozone_geo)
```

```
## variog: computing omnidirectional variogram
```

```
fit_exp <- variofit(
  var_classical,
  cov.model = "exponential",
  ini.cov.pars = c(0.00012, 10),
  nugget = 0.00001
)
```

```
## variofit: covariance model used is exponential
## variofit: weights used: npairs
## variofit: minimisation function used: optim
```

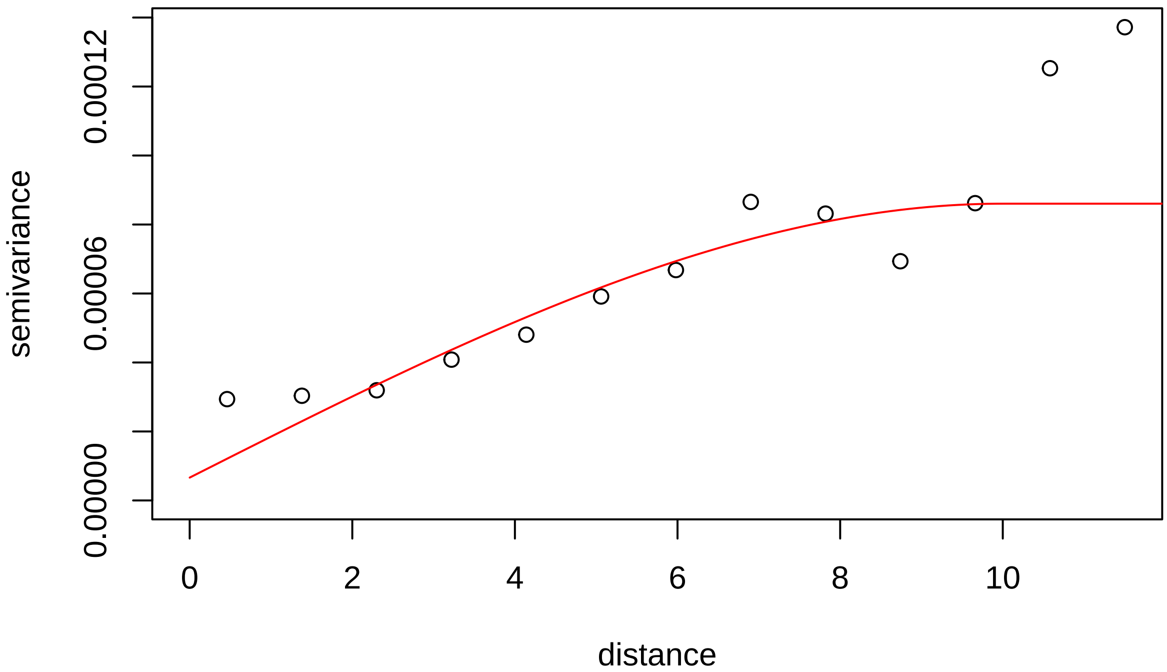
```
plot(var_classical)
lines(fit_exp,col="red")
```




```r
fit_sph <- variofit(
  var_classical,
  cov.model = "spherical",
  ini.cov.pars = c(0.00012, 10),
  nugget = 0.00001
)
```

```
## variofit: covariance model used is spherical
## variofit: weights used: npairs
## variofit: minimisation function used: optim
```

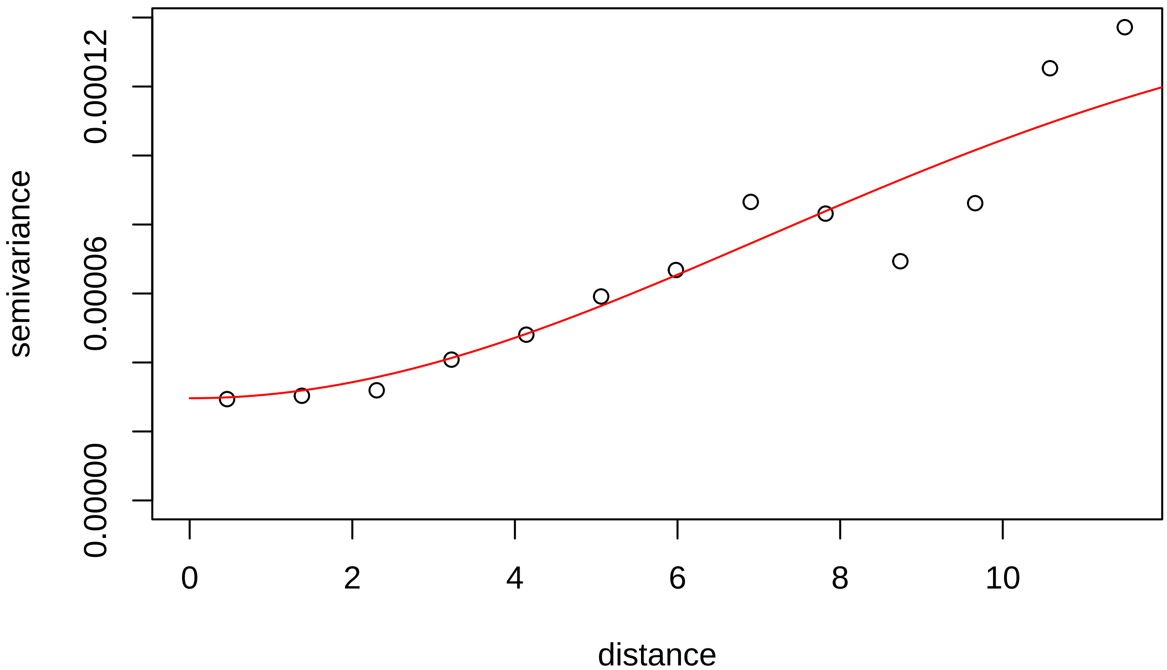
```r
plot(var_classical)
lines(fit_sph,col="red")
```




```
fit_gau <- variofit(
  var_classical,
  cov.model = "gaussian",
  ini.cov.pars = c(0.00012, 10),
  nugget = 0.00003
)
```

```
## variofit: covariance model used is gaussian
## variofit: weights used: npairs
## variofit: minimisation function used: optim
```

```
plot(var_classical)
lines(fit_gau,col="red")
```



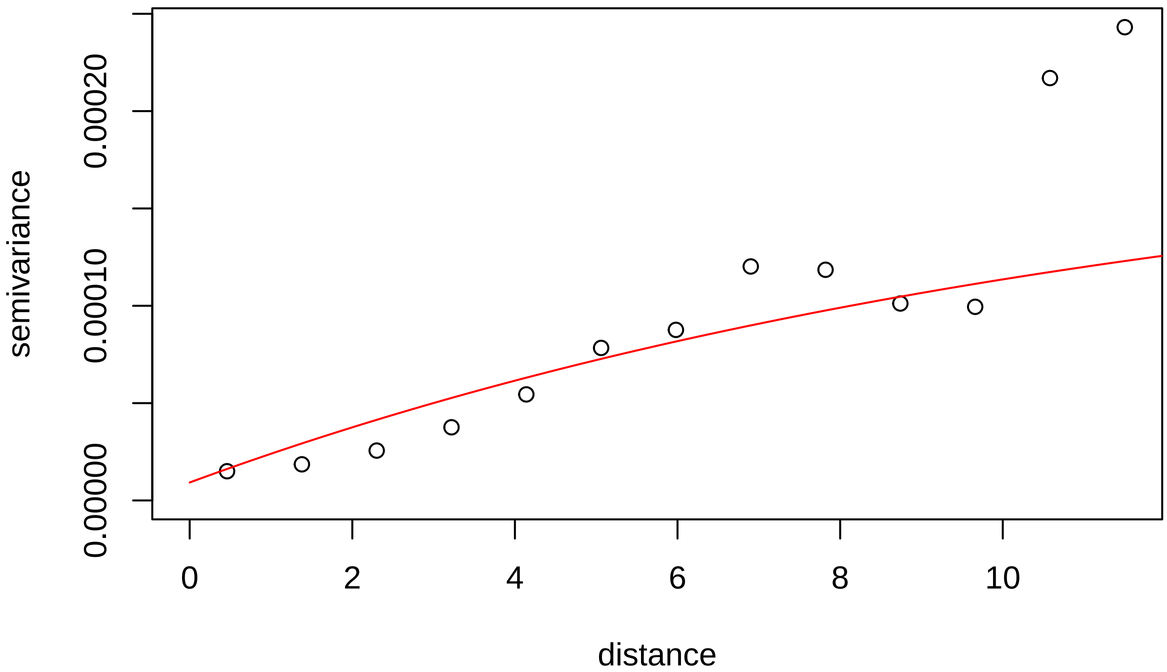

```r
var_robust <- variog(ozone_geo, estimator.type = "modulus")
```

```
## variog: computing omnidirectional variogram
```

```r
fit_exp2 <- variofit(
  var_robust,
  cov.model = "exponential",
  ini.cov.pars = c(0.0002, 12),
  nugget = 0.00001
)
```

```
## variofit: covariance model used is exponential
## variofit: weights used: npairs
## variofit: minimisation function used: optim
```

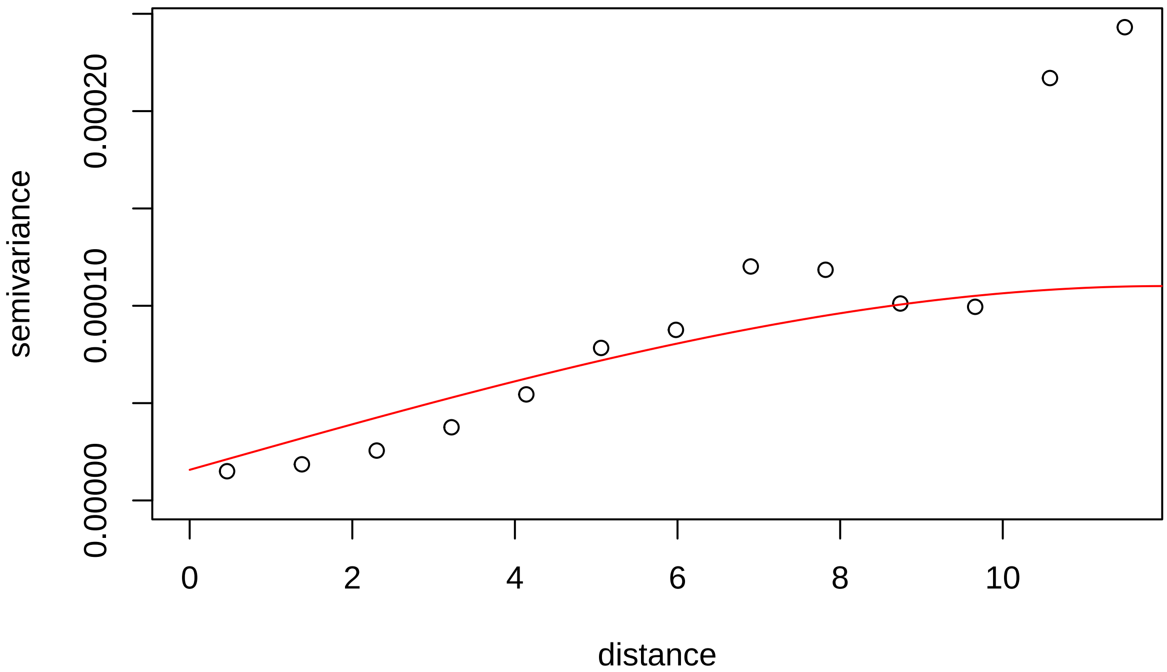
```r
plot(var_robust)
lines(fit_exp2,col="red")
```




```r
fit_sph2 <- variofit(
  var_robust,
  cov.model = "spherical",
  ini.cov.pars = c(0.00012, 12),
  nugget = 0.00002
)
```

```
## variofit: covariance model used is spherical
## variofit: weights used: npairs
## variofit: minimisation function used: optim
```

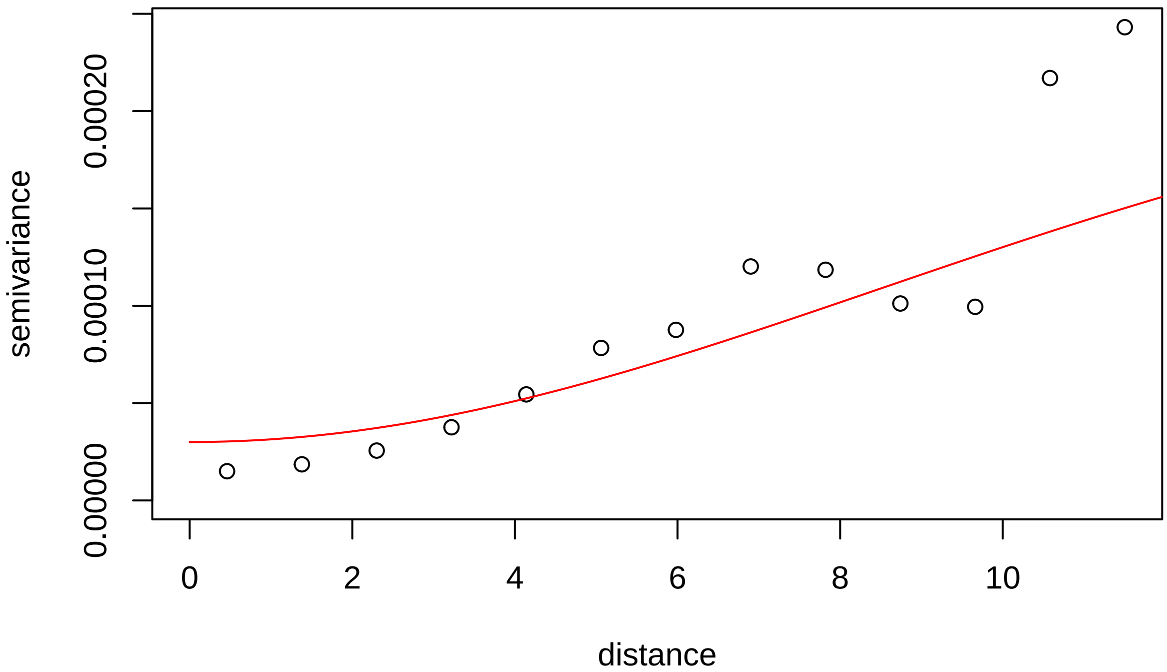
```r
plot(var_robust)
lines(fit_sph2,col="red")
```




```r
fit_gau2 <- variofit(
  var_robust,
  cov.model = "gaussian",
  ini.cov.pars = c(0.0002, 12),
  nugget = 0.00003
)
```

```
## variofit: covariance model used is gaussian
## variofit: weights used: npairs
## variofit: minimisation function used: optim
```

```r
plot(var_robust)
lines(fit_gau2,col="red")
```



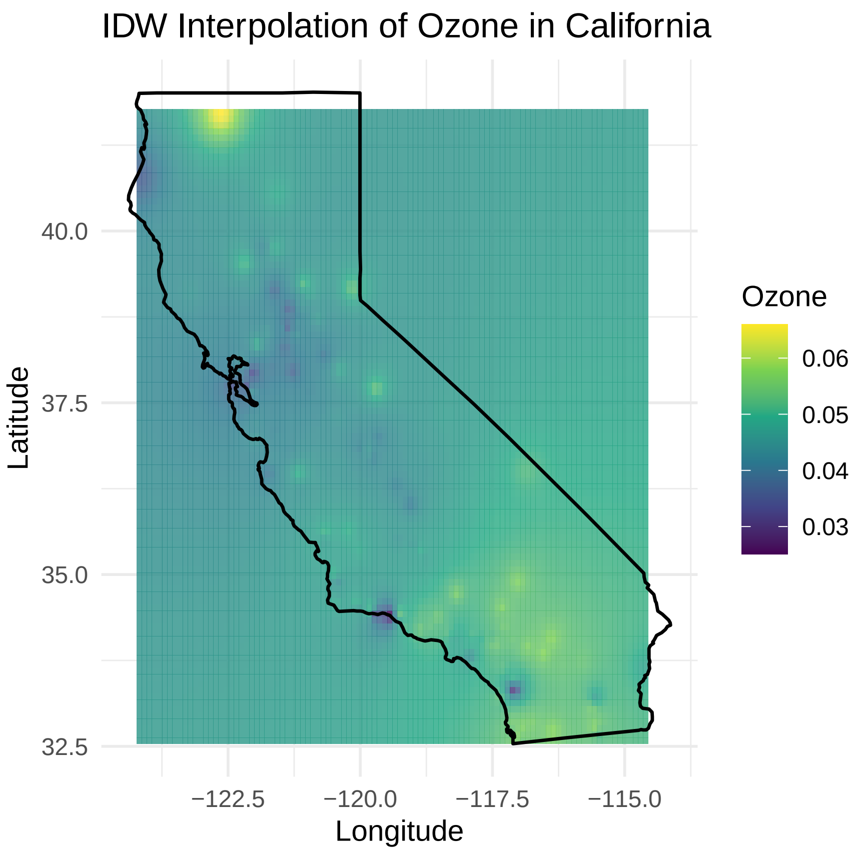
Dense Grid

```
x.range <- seq(min(relevant$Site.Longitude),
               max(relevant$Site.Longitude),
               length = 100)

y.range <- seq(min(relevant$Site.Latitude),
               max(relevant$Site.Latitude),
               length = 100)

grid <- expand.grid(
  Site.Longitude = x.range,
  Site.Latitude = y.range
)

coordinates(relevant) <- ~Site.Longitude + Site.Latitude
coordinates(grid) <- ~Site.Longitude + Site.Latitude
gridded(grid) <- TRUE

idw_pred <- idw(
  Daily.Max.8.hour.Ozone.Concentration ~ 1,
  relevant,
  newdata = grid,
  idp = 2
)
```

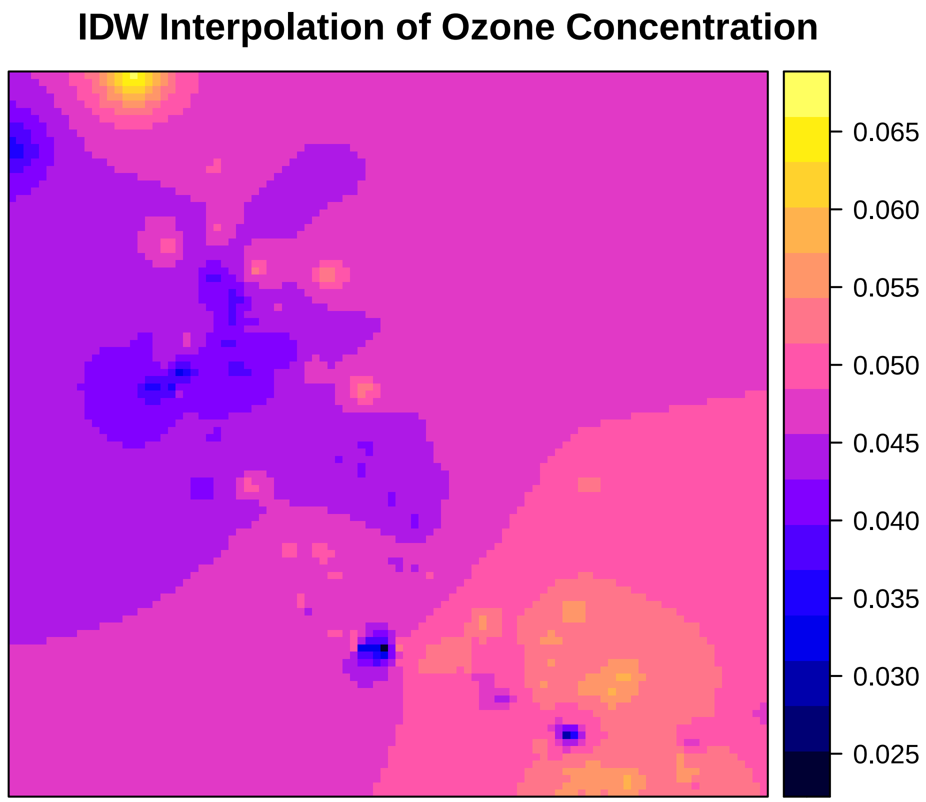
```
## [inverse distance weighted interpolation]
```

```
idw_df <- as.data.frame(idw_pred)

ggplot() +
  geom_tile(data = idw_df,
```

```
           aes(x = Site.Longitude,
               y = Site.Latitude,
               fill = var1.pred),
           alpha = 0.8) +
  geom_polygon(data = california,
              aes(x = long, y = lat, group = group),
              fill = NA,
              color = "black",
              linewidth = 0.6) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ozone") +
  labs(title = "IDW Interpolation of Ozone in California",
       x = "Longitude",
       y = "Latitude") +
  theme_minimal()
```




```
spplot(idw_pred["var1.pred"],
       main = "IDW Interpolation of Ozone Concentration")
```



Ordinary Kriging

```
x.range <- seq(min(relevant$Site.Longitude),
               max(relevant$Site.Longitude),
               length=100)

y.range <- seq(min(relevant$Site.Latitude),
               max(relevant$Site.Latitude),
               length=100)

grid <- expand.grid(x=x.range, y=y.range)

ok_pred <- krige.conv(
  ozone_geo,
  locations = grid,
  krige = krige.control(
    type.krige = "ok",
    cov.model = fit_exp$cov.model,
    cov.pars = fit_exp$cov.pars,
    nugget = fit_exp$nugget
  )
)
```

```
## krige.conv: model with constant mean
## krige.conv: Kriging performed using global neighbourhood
```

```
sk_pred <- krige.conv(
  ozone_geo,
```

```r
  locations = grid,
  krige = krige.control(
    type.krige = "sk",
    beta = mean(ozone_geo$data),
    cov.model = fit_exp$cov.model,
    cov.pars = fit_exp$cov.pars,
    nugget = fit_exp$nugget
  )
)
```

```
## krige.conv: model with constant mean
## krige.conv: Kriging performed using global neighbourhood
```

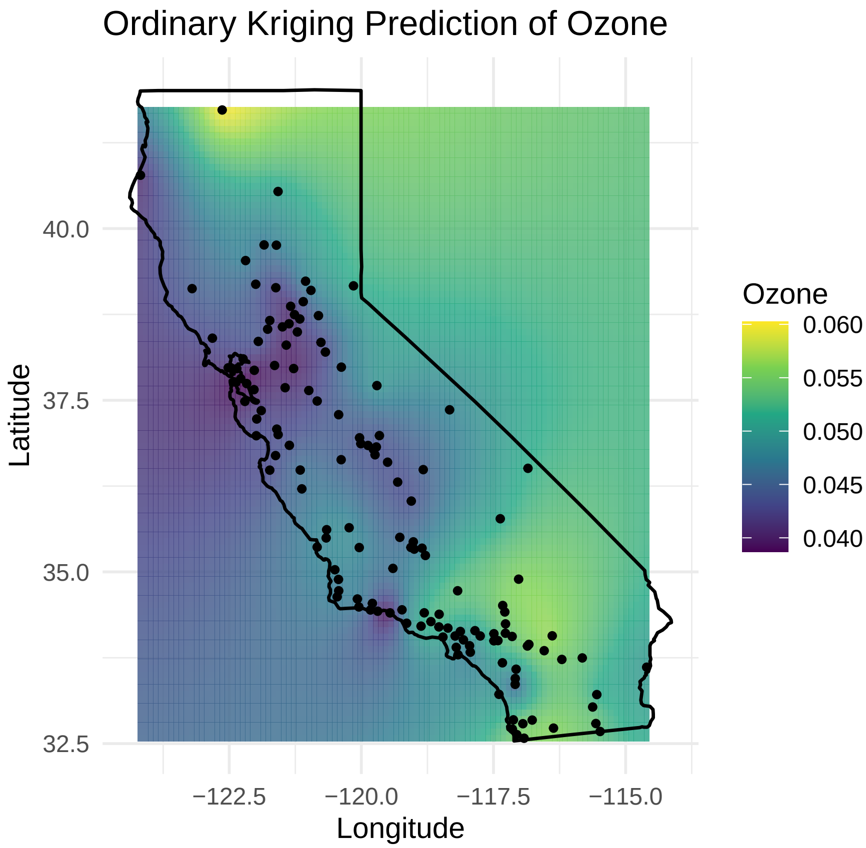
```r
uk_pred <- krige.conv(
  ozone_geo,
  locations = grid,
  krige = krige.control(
    type.krige = "ok",
    trend.d = "1st",
    trend.l = "1st",
    cov.model = fit_exp$cov.model,
    cov.pars = fit_exp$cov.pars,
    nugget = fit_exp$nugget
  )
)
```

```
## krige.conv: model with mean given by a 1st order polynomial on the coordinates
## krige.conv: Kriging performed using global neighbourhood
```

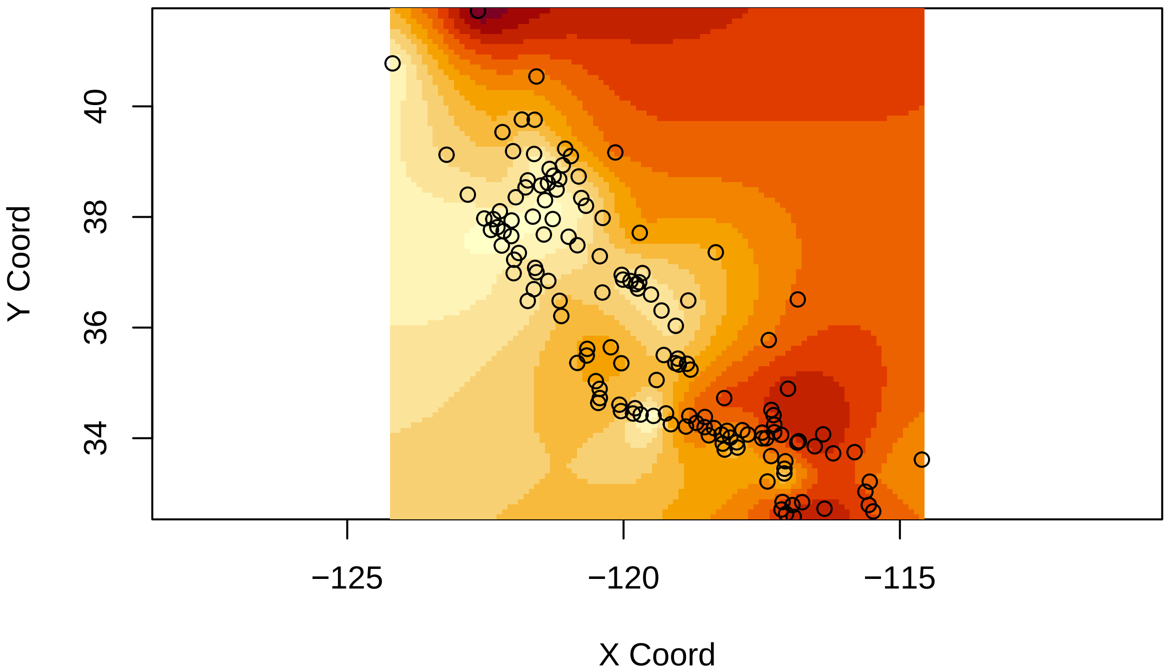
Ordinary Kriging

```r
krig_df <- data.frame(
  x = grid$x,
  y = grid$y,
  pred = ok_pred$predict
)

relevant_df <- as.data.frame(relevant)

ggplot() +
  geom_tile(data = krig_df,
            aes(x = x, y = y, fill = pred),
            alpha = 0.8) +
  geom_polygon(data = california,
               aes(x = long, y = lat, group = group),
               fill = NA,
               color = "black",
               linewidth = 0.6) +
  geom_point(data = relevant_df,
             aes(x = Site.Longitude,
                 y = Site.Latitude),
             color = "black",
             size = 1) +
  coord_fixed(1.3) +
```

```
  scale_fill_viridis_c(name = "Ozone") +
  labs(title = "Ordinary Kriging Prediction of Ozone",
       x = "Longitude",
       y = "Latitude") +
  theme_minimal()
```

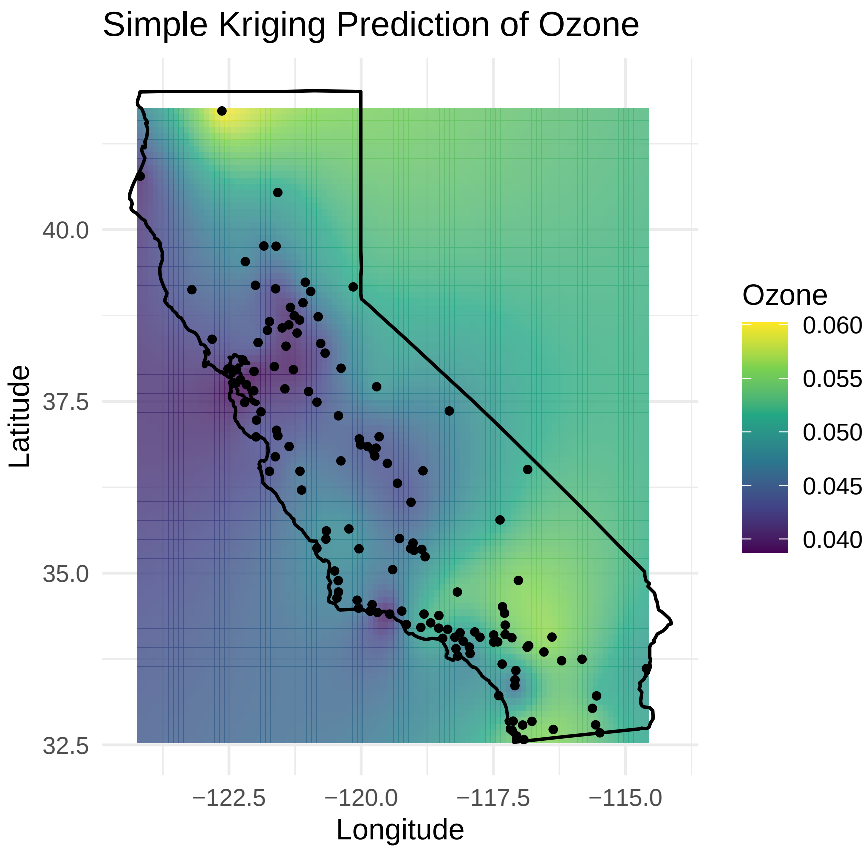



```
image(ok_pred)
points(ozone_geo$coords)
```

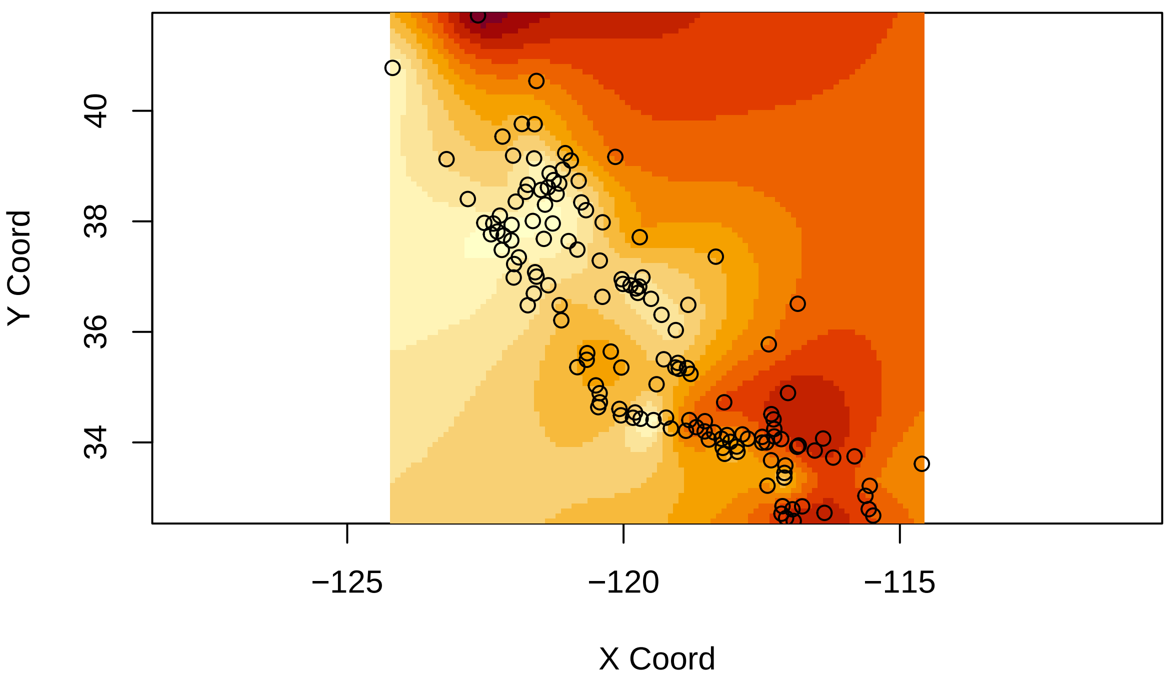


Simple Kriging

```r
krig_df <- data.frame(
  x = grid$x,
  y = grid$y,
  pred = sk_pred$predict
)

relevant_df <- as.data.frame(relevant)

ggplot() +
  geom_tile(data = krig_df,
            aes(x = x, y = y, fill = pred),
            alpha = 0.8) +
  geom_polygon(data = california,
               aes(x = long, y = lat, group = group),
               fill = NA,
               color = "black",
               linewidth = 0.6) +
  geom_point(data = relevant_df,
             aes(x = Site.Longitude,
                 y = Site.Latitude),
             color = "black",
             size = 1) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ozone") +
  labs(title = "Simple Kriging Prediction of Ozone",
       x = "Longitude",
       y = "Latitude") +
  theme_minimal()
```

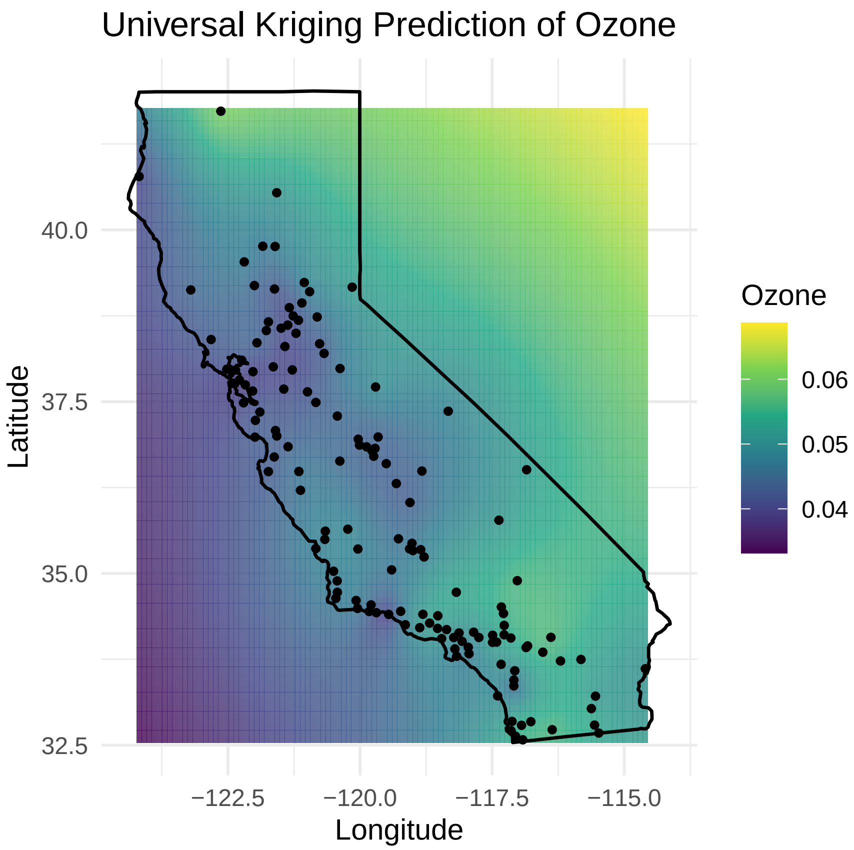



```r
image(sk_pred)
points(ozone_geo$coords)
```

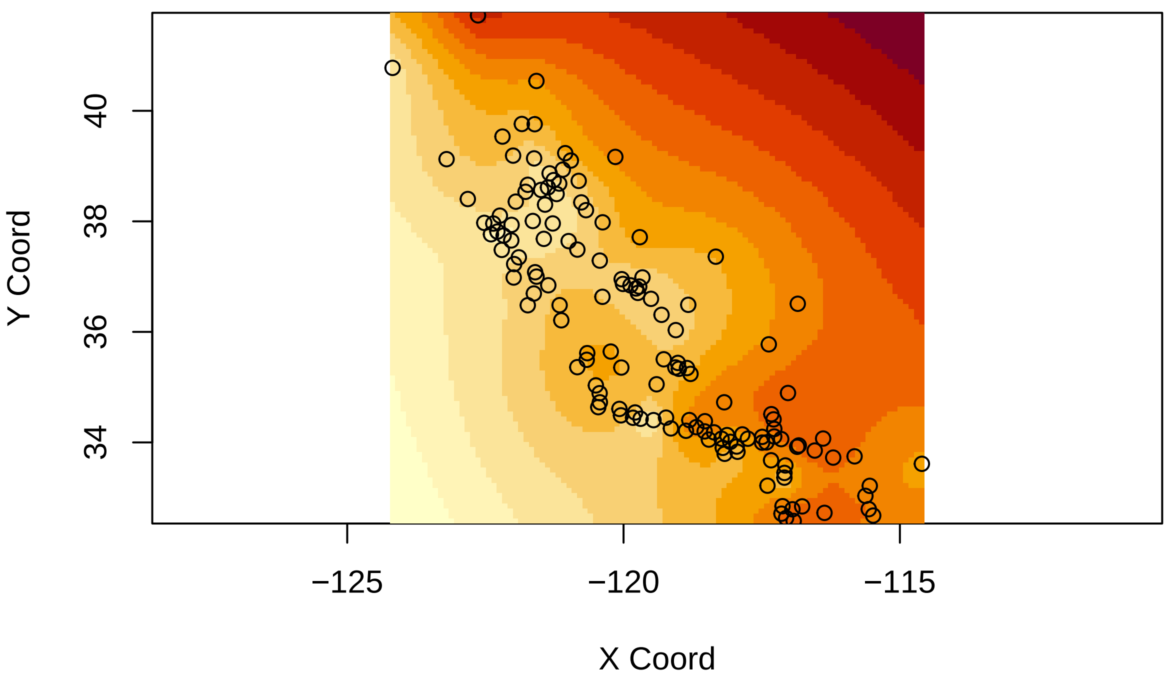


Universal Kriging

```r
krig_df <- data.frame(
  x = grid$x,
  y = grid$y,
  pred = uk_pred$predict
)

relevant_df <- as.data.frame(relevant)

ggplot() +
  geom_tile(data = krig_df,
            aes(x = x, y = y, fill = pred),
            alpha = 0.8) +
  geom_polygon(data = california,
               aes(x = long, y = lat, group = group),
               fill = NA,
               color = "black",
               linewidth = 0.6) +
  geom_point(data = relevant_df,
             aes(x = Site.Longitude,
                 y = Site.Latitude),
             color = "black",
             size = 1) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ozone") +
  labs(title = "Universal Kriging Prediction of Ozone",
       x = "Longitude",
       y = "Latitude") +
  theme_minimal()
```




```r
image(uk_pred)
points(ozone_geo$coords)
```



Cross Validation

```
cv_exp <- xvalid(
  ozone_geo,
  model=fit_exp
)
```

```
## xvalid: number of data locations       = 142
## xvalid: number of validation locations = 142
## xvalid: performing cross-validation at location ... 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1
## xvalid: end of cross-validation
```

```
cv_sph <- xvalid(
  ozone_geo,
  model=fit_sph
)
```

```
## xvalid: number of data locations       = 142
## xvalid: number of validation locations = 142
## xvalid: performing cross-validation at location ... 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1
## xvalid: end of cross-validation
```

```
cv_gau <- xvalid(
  ozone_geo,
  model=fit_gau
)
```

```
## xvalid: number of data locations       = 142
## xvalid: number of validation locations = 142
## xvalid: performing cross-validation at location ... 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1
## xvalid: end of cross-validation
```

```
cv_ok <- xvalid(
  ozone_geo,
  model = fit_exp,
  krige = krige.control(type.krige = "ok")
)
```

```
## xvalid: number of data locations       = 142
## xvalid: number of validation locations = 142
## xvalid: performing cross-validation at location ... 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1
## xvalid: end of cross-validation
```

```
cv_sk <- xvalid(
  ozone_geo,
  model = fit_exp,
  krige = krige.control(
    type.krige = "sk",
    beta = mean(ozone_geo$data)
  )
)
```

```
## xvalid: number of data locations       = 142
## xvalid: number of validation locations = 142
## xvalid: performing cross-validation at location ... 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 1
## xvalid: end of cross-validation
```

```r
cv_uk <- xvalid(
  ozone_geo,
  model = fit_exp,
  krige = krige.control(
    type.krige = "ok",
    trend.d = "1st",
    trend.l = "1st"
  )
)
```

```
## xvalid: number of data locations       = 142
## xvalid: number of validation locations = 142
## xvalid: performing cross-validation at location ... 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
## xvalid: end of cross-validation
```

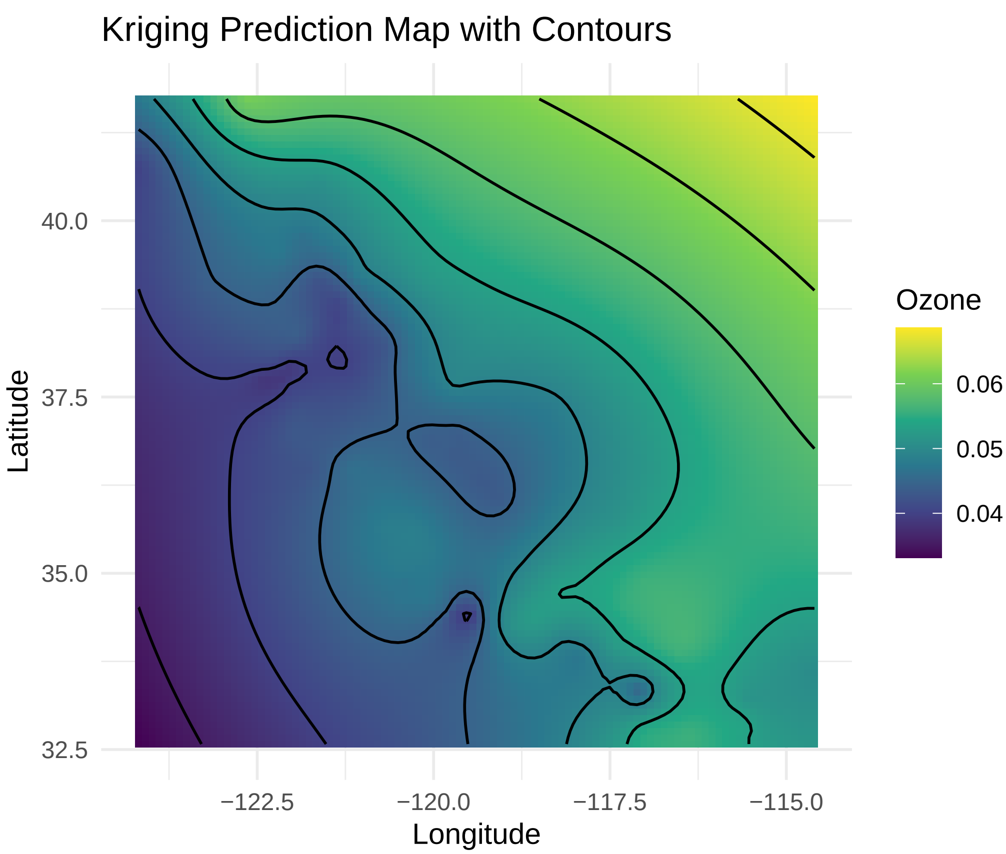
```r
ME_exp  <- mean(cv_exp$error)
MSE_exp <- mean(cv_exp$error^2)

ME_sph  <- mean(cv_sph$error)
MSE_sph <- mean(cv_sph$error^2)

ME_gau  <- mean(cv_gau$error)
MSE_gau <- mean(cv_gau$error^2)

ME_ok  <- mean(cv_ok$error)
MSE_ok <- mean(cv_ok$error^2)

ME_sk  <- mean(cv_sk$error)
MSE_sk <- mean(cv_sk$error^2)

ME_uk  <- mean(cv_uk$error)
MSE_uk <- mean(cv_uk$error^2)

results <- data.frame(
  Model=c("Exponential","Spherical","Gaussian","Ordinary Krig", "Simple Krig", "Universal Krig"),
  ME=c(ME_exp,ME_sph,ME_gau,ME_ok,ME_sk,ME_uk ),
  MSE=c(MSE_exp,MSE_sph,MSE_gau,MSE_ok,MSE_sk,MSE_uk)
)

results
```

```
##            Model           ME          MSE
## 1    Exponential 3.171732e-05 3.034311e-05
## 2      Spherical 3.117738e-05 3.054154e-05
## 3       Gaussian 1.302923e-05 3.235068e-05
## 4  Ordinary Krig 3.171732e-05 3.034311e-05
## 5    Simple Krig 3.171732e-05 3.034311e-05
## 6 Universal Krig 3.171732e-05 3.034311e-05
```

Universal Krig Raster Map

```r
ggplot() +
  geom_tile(data = krig_df,
            aes(x = x, y = y, fill = pred)) +
```

```
  geom_contour(data = krig_df,
               aes(x = x, y = y, z = pred),
               color = "black",
               bins = 10) +
  coord_fixed() +
  scale_fill_viridis_c(name = "Ozone") +
  labs(title = "Kriging Prediction Map with Contours",
       x = "Longitude",
       y = "Latitude") +
  theme_minimal()
```




```
ggplot() +
  geom_tile(data = krig_df,
            aes(x = x, y = y, fill = pred),
            alpha = 0.9) +
  geom_contour(data = krig_df,
               aes(x = x, y = y, z = pred),
               color = "black",
               bins = 10) +
  geom_polygon(data = california,
               aes(x = long, y = lat, group = group),
               fill = NA,
               color = "red",
               linewidth = 0.6) +
  coord_fixed(1.3) +
  scale_fill_viridis_c(name = "Ozone") +
  labs(title = "Kriging Prediction Surface with Contours",
```

```
      x = "Longitude",
      y = "Latitude") +
  theme_minimal()
```